Digit RBKD

1.0

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1 Main Page 1

1.1 Digit_RBKD . . . 1

1.2 Work in Progress . . . 1

2 Hierarchical Index 3

2.1 Class Hierarchy . . . 3

3 Class Index 5

3.1 Class List . . . 5

4 Class Documentation 7

4.1 Eigen::internal::cast_impl< casadi::SX, Scalar > Struct Template Reference . . . 7

4.1.1 Detailed Description . . . 7

4.2 DigitModel Class Reference . . . 7

4.2.1 Detailed Description . . . 8

4.3 CasadiEigenCodeGen::DynamicSizeCG Class Reference . . . 8

4.3.1 Detailed Description . . . 9

4.3.2 Member Function Documentation . . . 9

4.3.2.1 add_casadi_function() . . . 9

4.4 rbda::Er_struct Struct Reference . . . 10

4.4.1 Detailed Description . . . 10

4.5 CasadiEigenCodeGen::FixedSizeCG Class Reference . . . 11

4.5.1 Detailed Description . . . 12

4.5.2 Member Function Documentation . . . 12

4.5.2.1 add_casadi_function() . . . 12

4.5.2.2 generate_eigen_function_signature() . . . 13

4.5.2.3 generate_eigen_function_source() . . . 14

4.5.2.4 get_casadi_function_io_dims() . . . 15

4.5.2.5 get_casadi_function_io_names() . . . 15

4.6 rbda::ForceVector Struct Reference . . . 16

4.6.1 Detailed Description . . . 17

4.7 Joint Struct Reference . . . 17

4.7.1 Detailed Description . . . 17

4.8 rbda::mcI_struct Struct Reference . . . 17

4.8.1 Detailed Description . . . 18

4.9 rbda::MotionSubspace Struct Reference . . . 18

4.9.1 Detailed Description . . . 18

4.10 rbda::MotionVector Struct Reference . . . 18

4.10.1 Detailed Description . . . 19

4.11 Eigen::NumTraits< casadi::SX > Struct Reference . . . 19

4.11.1 Detailed Description . . . 20

4.12 rbda::PluckerTransform Struct Reference . . . 20

4.12.1 Detailed Description . . . 20

4.13 rbda::Pose Struct Reference 21
4.13.1 Detailed Description 21
4.14 RigidBody Struct Reference 21
4.14.1 Detailed Description 22
4.15 RigidBodyTree Class Reference 22
4.15.1 Detailed Description 23
4.15.2 Member Function Documentation 24
4.15.2.1 kappa() 24
4.15.2.2 print_state() 24
4.16 Site Struct Reference 25
4.16.1 Detailed Description 25
4.17 rbda::SpatialInertia Struct Reference 26
4.17.1 Detailed Description 26

Index **27**

Chapter 1

Main Page



1.1 Digit_RBKD

Rigid Body Kinematics and Dynamics in c++ for robot Digit from Agility Robotics.

Why not Pinocchio or RBDL? This way I gain more understanding of the underlying `algorithms` while making it specifically suited for my control code down the line.

This implementation is 9.7 times slower than Pinocchio but has pedagogical value

Please star if you are using it

1.2 Work in Progress

Chapter 2

Hierarchical Index

2.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

Eigen::internal::cast_impl< casadi::SX, Scalar > . . . 7
CodeGenerator
  CasadiEigenCodeGen::FixedSizeCG . . . 11
    CasadiEigenCodeGen::DynamicSizeCG . . . 8
rbda::Er_struct . . . 10
rbda::ForceVector . . . 16
Joint . . . 17
rbda::mcl_struct . . . 17
rbda::MotionSubspace . . . 18
rbda::MotionVector . . . 18
Eigen::NumTraits< casadi::SX > . . . 19
rbda::PluckerTransform . . . 20
rbda::Pose . . . 21
RigidBody . . . 21
RigidBodyTree . . . 22
  DigitModel . . . 7
Site . . . 25
rbda::SpatialInertia . . . 26

Chapter 3

Class Index

3.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

Eigen::internal::cast_impl< casadi::SX, Scalar > 7
DigitModel
    Class for the DigitModel 7
CasadiEigenCodeGen::DynamicSizeCG 8
rbda::Er_struct
    A struct to hold E and r 10
CasadiEigenCodeGen::FixedSizeCG
    The Helper class for code generation for eigen 11
rbda::ForceVector 16
Joint 17
rbda::mcI_struct
    Struct to hold mass (1x1), center of mass location (3x1), and inertia matrix (3x3) 17
rbda::MotionSubspace 18
rbda::MotionVector 18
Eigen::NumTraits< casadi::SX > 19
rbda::PluckerTransform 20
rbda::Pose 21
RigidBody 21
RigidBodyTree 22
Site 25
rbda::SpatialInertia 26

Chapter 4

Class Documentation

4.1 Eigen::internal::cast_impl< casadi::SX, Scalar > Struct Template Reference

Static Public Member Functions

- static Scalar **run** (const casadi::SX &x)

4.1.1 Detailed Description

template<typename Scalar>
struct Eigen::internal::cast_impl< casadi::SX, Scalar >

Definition at line 47 of file common_defines.hxx.

The documentation for this struct was generated from the following file:

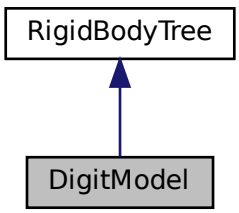
- /home/abhijee/Documents/GitHub/Digit_RBKD/include/digit_rbkd/common_defines.hxx

4.2 DigitModel Class Reference

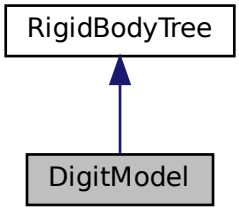
Class for the DigitModel.

`#include <digit_model_library.hpp>`

Inheritance diagram for DigitModel:



Collaboration diagram for DigitModel:



Additional Inherited Members

4.2.1 Detailed Description

Class for the DigitModel.

Created after mujoco model of digit.

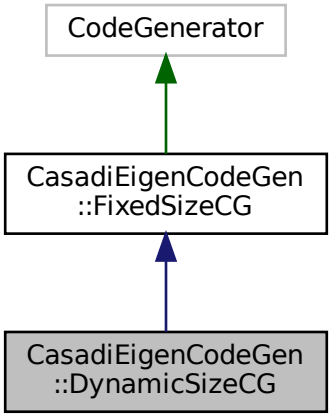
Definition at line 85 of file digit_model_library.hpp.

The documentation for this class was generated from the following files:

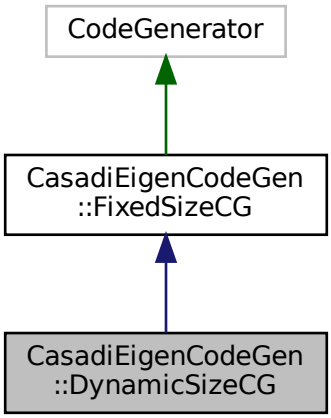
- /home/abhijee/Documents/GitHub/Digit_RBKD/include/digit_rbkd/digit_model_library.hpp
- /home/abhijee/Documents/GitHub/Digit_RBKD/src/digit_model_library.cpp

4.3 CasadiEigenCodeGen::DynamicSizeCG Class Reference

Inheritance diagram for CasadiEigenCodeGen::DynamicSizeCG:



Collaboration diagram for CasadiEigenCodeGen::DynamicSizeCG:



Public Member Functions

- **DynamicSizeCG** (const std::string &name, const std::string &code_gen_dir, const Dict &opts)
- void add_casadi_function (const Function &f)

  Adds a casadi function to the eigen code generator.

Additional Inherited Members

4.3.1 Detailed Description

Definition at line 257 of file CasadiEigenCodeGenerator.hpp.

4.3.2 Member Function Documentation

4.3.2.1 add_casadi_function()

```
void CasadiEigenCodeGen::DynamicSizeCG::add_casadi_function (
            const Function & f )  [inline]
```

Adds a casadi function to the eigen code generator.

Parameters

f	casadi function

Definition at line 267 of file CasadiEigenCodeGenerator.hpp.

```
267                                                 {
268             // Add the function to the code generator
269             this->add(f);
270 
271             // Get argument dimensions and names
272             std::pair<std::vector<std::pair<int, int>, std::vector<std::pair<int, int>>> dims =
    get_casadi_function_io_dims(
273                     f);
274             std::pair<std::vector<std::string>, std::vector<std::string>> names =
    get_casadi_function_io_names(f);
275             std::vector<std::pair<int, int>> argument_dims = dims.first;
276             std::vector<std::pair<int, int>> result_dims = dims.second;
277             std::vector<std::string> argument_names = names.first;
278             std::vector<std::string> result_names = names.second;
279             std::string function_name = f.name();
280 
281             // Add the eigen function signature to the header
282             this->declare(generate_eigen_function_signature(argument_names, result_names,
    function_name));
283             // Add the eigen function implementation to the source
284             this->body << generate_eigen_function_source(argument_names, result_names, function_name);
285         }
```

The documentation for this class was generated from the following file:

- /home/abhijee/Documents/GitHub/Digit_RBKD/include/digit_rbkd/CasadiEigenCodeGenerator.hpp

4.4 rbda::Er_struct Struct Reference

A struct to hold E and r.

`#include <spatial_vector_arithmetic.hxx>`

Public Attributes

- Eigen::Matrix< myfloat, 3, 3 > **E**
- Eigen::Matrix< myfloat, 3, 1 > **r**

4.4.1 Detailed Description

A struct to hold E and r.

Definition at line 195 of file spatial_vector_arithmetic.hxx.

The documentation for this struct was generated from the following file:

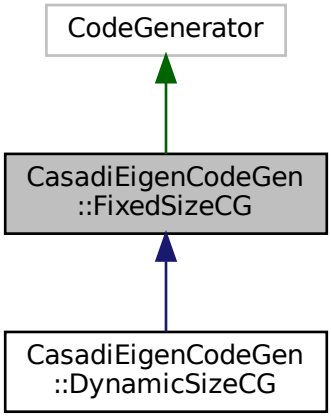
- /home/abhijee/Documents/GitHub/Digit_RBKD/include/digit_rbkd/rbda/spatial_vector_arithmetic.hxx

4.5 CasadiEigenCodeGen::FixedSizeCG Class Reference

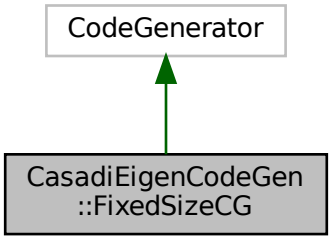
The Helper class for code generation for eigen.

`#include <CasadiEigenCodeGenerator.hpp>`

Inheritance diagram for CasadiEigenCodeGen::FixedSizeCG:



Collaboration diagram for CasadiEigenCodeGen::FixedSizeCG:



Public Member Functions

- **FixedSizeCG** (const std::string &name, const std::string &code_gen_dir, const Dict &opts)
- void generate_with_eigen ()

  Generates the code using the eigen code generator.
- void add_casadi_function (const Function &f)

  Adds a casadi function to the eigen code generator.

Protected Member Functions

- std::pair< std::vector< std::pair< int, int > >, std::vector< std::pair< int, int > > > get_casadi_function_io_dims (const Function &f, bool print=false)

  Returns the pair of vectors of pairs containing input and output dimensions of the casadi function.
- std::pair< std::vector< std::string >, std::vector< std::string > > get_casadi_function_io_names (const Function &f, bool print=false)

  Returns pair of vectors containing input and output names of the casadi function.
- std::string generate_eigen_function_signature (const std::vector< std::pair< int, int >> &argument_dims, const std::vector< std::pair< int, int >> &result_dims, const std::vector< std::string > &argument_names, const std::vector< std::string > &result_names, const std::string &function_name)

  Returns the Eigen function signature.
- std::string generate_eigen_function_source (const std::vector< std::pair< int, int >> &argument_dims, const std::vector< std::pair< int, int >> &result_dims, const std::vector< std::string > &argument_names, const std::vector< std::string > &result_names, const std::string &function_name)

  Returns the Eigen function implementation.

Protected Attributes

- std::string **code_gen_dir**

4.5.1 Detailed Description

The Helper class for code generation for eigen.

Definition at line 37 of file CasadiEigenCodeGenerator.hpp.

4.5.2 Member Function Documentation

4.5.2.1 add_casadi_function()

```
void CasadiEigenCodeGen::FixedSizeCG::add_casadi_function (
            const Function & f )  [inline]
```

Adds a casadi function to the eigen code generator.

Parameters

f	casadi function

Definition at line 85 of file CasadiEigenCodeGenerator.hpp.

```
85                                                  {
86             // Add the function to the code generator
87             this->add(f);
88
89             // Get argument dimensions and names
90             std::pair<std::vector<std::pair<int, int», std::vector<std::pair<int, int»> dims =
       get_casadi_function_io_dims(
```

```
91                  f);
92          std::pair<std::vector<std::string>, std::vector<std::string» names =
    get_casadi_function_io_names(f);
93          std::vector<std::pair<int, int» argument_dims = dims.first;
94          std::vector<std::pair<int, int» result_dims = dims.second;
95          std::vector<std::string> argument_names = names.first;
96          std::vector<std::string> result_names = names.second;
97          std::string function_name = f.name();
98
99          // Add the eigen function signature to the header
100          this->declare(generate_eigen_function_signature(argument_dims, result_dims, argument_names,
    result_names,
101                                                          function_name));
102          // Add the eigen function implementation to the source
103          this->body « generate_eigen_function_source(argument_dims, result_dims, argument_names,
    result_names,
104                                                      function_name);
105      }
```

4.5.2.2 generate_eigen_function_signature()

```
std::string CasadiEigenCodeGen::FixedSizeCG::generate_eigen_function_signature (
            const std::vector< std::pair< int, int >> & argument_dims,
            const std::vector< std::pair< int, int >> & result_dims,
            const std::vector< std::string > & argument_names,
            const std::string & function_name )  [inline], [protected]
```

Returns the Eigen function signature.

Parameters

argument_dims	a pair of the form {rows,cols} for each argument
result_dims	a pair of the form {rows,cols} for each result
argument_names	a vector of argument names
result_names	a vector of result names
function_name	name of the function

Returns

Eigen function implementation

Definition at line 180 of file CasadiEigenCodeGenerator.hpp.

```
184                                                                                                 {
185          // create a void function signature
186          // with arguments as references to Eigen matrices with appropriate dimensions
187
188          std::string function = "void " + function_name + "_eigen(";
189          for (int i = 0; i < argument_dims.size(); ++i) {
190              if (i != 0) {
191                  function += "\t\t\t\t";
192              }
193              function += "const Eigen::Matrix<casadi_real," + std::to_string(argument_dims[i].first)
    + "," +
194                          std::to_string(argument_dims[i].second) + "> &" + argument_names[i];
195              function += ",\n";
196          }
197          for (int i = 0; i < result_dims.size(); ++i) {
198              function += "\t\t\t\tEigen::Matrix<casadi_real," + std::to_string(result_dims[i].first)
    + "," +
199                          std::to_string(result_dims[i].second) + "> &" + result_names[i];
200              if (i < result_dims.size() - 1) {
201                  function += ", \n";
202              }
```

```
203             }
204             function += ")";
205 
206             return function;
207 
208         }
```

4.5.2.3 generate_eigen_function_source()

```
std::string CasadiEigenCodeGen::FixedSizeCG::generate_eigen_function_source (
            const std::vector< std::pair< int, int >> & argument_dims,
            const std::vector< std::pair< int, int >> & result_dims,
            const std::vector< std::string > & argument_names,
            const std::vector< std::string > & result_names,
            const std::string & function_name )  [inline], [protected]
```

Returns the Eigen function implementation.

Parameters

argument_dims	a pair of the form {rows,cols} for each argument
result_dims	a pair of the form {rows,cols} for each result
argument_names	a vector of argument names
result_names	a vector of result names
function_name	name of the function

Returns

Eigen function implementation

Definition at line 218 of file CasadiEigenCodeGenerator.hpp.

```
222                                                                                 {
223             // create a void function implementation
224             std::string source = "extern \"C\" CASADI_SYMBOL_EXPORT ";
225             source += generate_eigen_function_signature(argument_dims,
226                                                         result_dims,
227                                                         argument_names,
228                                                         result_names,
229                                                         function_name);
230             source += " {\n";
231             source += "// create arguments for generated casadi function\n";
232             source += "const casadi_real *arg[] = {";
233             for (int i = 0; i < argument_names.size(); ++i) {
234                 source += argument_names[i] + ".data()";
235                 if (i < argument_names.size() - 1) {
236                     source += ", ";
237                 }
238             }
239             source += "};\n";
240             source += "// create results for generated casadi function\n";
241             source += "casadi_real *res[] = {";
242             for (int i = 0; i < result_names.size(); ++i) {
243                 source += result_names[i] + ".data()";
244                 if (i < result_names.size() - 1) {
245                     source += ", ";
246                 }
247             }
248             source += "};\n";
249             source += "// call the generated casadi function\n";
250             source += function_name + "(arg, res, nullptr, nullptr, 0);\n";
251             source += "}\n";
252             return source;
253         }
```

4.5.2.4 get_casadi_function_io_dims()

```
std::pair<std::vector<std::pair<int, int> >, std::vector<std::pair<int, int> > > CasadiEigenCodeGen::FixedSizeCG::get_casadi_function_io_dims (
            const Function & f,
            bool print = false )  [inline], [protected]
```

Returns the pair of vectors of pairs containing input and output dimensions of the casadi function.

Parameters

f	casadi function
print	if true, prints the input and output dimensions

Returns

{[{rows:cols},{rows:cols}....] : [{rows:cols},{rows:cols}....]}

Definition at line 118 of file CasadiEigenCodeGenerator.hpp.

```
118                                                                                 {
119             const casadi_int num_args = f.sz_arg();
120             const casadi_int num_res = f.sz_res();
121             std::vector<std::pair<int, int» in_dims;
122             std::vector<std::pair<int, int» res_dims;
123             for (casadi_int i = 0; i < num_args; ++i) {
124                 in_dims.push_back({f.sparsity_in(i).rows(), f.sparsity_in(i).columns()});
125             }
126             for (casadi_int i = 0; i < num_res; ++i) {
127                 res_dims.push_back({f.sparsity_out(i).rows(), f.sparsity_out(i).columns()});
128             }
129             if (print) {
130                 std::cout « "Input dimensions: " « std::endl;
131                 for (auto &in_dim: in_dims) {
132                     std::cout « in_dim.first « "x" « in_dim.second « std::endl;
133                 }
134                 std::cout « "Output dimensions: " « std::endl;
135                 for (auto &res_dim: res_dims) {
136                     std::cout « res_dim.first « "x" « res_dim.second « std::endl;
137                 }
138             }
139             return {in_dims, res_dims};
140         }
```

4.5.2.5 get_casadi_function_io_names()

```
std::pair<std::vector<std::string>, std::vector<std::string> > CasadiEigenCodeGen::FixedSizeCG::get_casadi_function_io_names (
            const Function & f,
            bool print = false )  [inline], [protected]
```

Returns pair of vectors containing input and output names of the casadi function.

Parameters

f	casadi function
print	if true, prints the input and output names

Returns

pair of vectors containing input and output names

Definition at line 148 of file CasadiEigenCodeGenerator.hpp.

```
148                                                                                 {
149             const casadi_int num_args = f.sz_arg();
150             const casadi_int num_res = f.sz_res();
151             std::vector<std::string> in_names;
152             std::vector<std::string> res_names;
153             for (casadi_int i = 0; i < num_args; ++i) {
154                 in_names.push_back(f.name_in(i));
155             }
156             for (casadi_int i = 0; i < num_res; ++i) {
157                 res_names.push_back(f.name_out(i));
158             }
159             if (print) {
160                 std::cout « "Input names: " « std::endl;
161                 for (auto &in_name: in_names) {
162                     std::cout « in_name « std::endl;
163                 }
164                 std::cout « "Output names: " « std::endl;
165                 for (auto &res_name: res_names) {
166                     std::cout « res_name « std::endl;
167                 }
168             }
169             return {in_names, res_names};
170         }
```

The documentation for this class was generated from the following file:

- /home/abhijee/Documents/GitHub/Digit_RBKD/include/digit_rbkd/CasadiEigenCodeGenerator.hpp

4.6 rbda::ForceVector Struct Reference

Public Member Functions

- **ForceVector** (Eigen::Matrix< myfloat, 3, 1 > n, Eigen::Matrix< myfloat, 3, 1 > f)
- **ForceVector** (Eigen::Matrix< myfloat, 6, 1 > fv)
- **ForceVector** (Eigen::Matrix< myfloat, 6,-1 > fv)
- Eigen::Matrix< myfloat, 6, 1 > **toVector** () const
- ForceVector & **operator=** (const ForceVector &rhs)

Public Attributes

- Eigen::Matrix< myfloat, 3, 1 > **n**
- Eigen::Matrix< myfloat, 3, 1 > **f**

Friends

- std::ostream & **operator**<< (std::ostream &os, const ForceVector &fv)
- ForceVector **operator+** (const ForceVector &lhs, const ForceVector &rhs)
- ForceVector **operator-** (const ForceVector &lhs, const ForceVector &rhs)
- ForceVector **operator**∗ (const myfloat &lhs, const ForceVector &rhs)
- ForceVector **operator**∗ (const ForceVector &lhs, const myfloat &rhs)

4.6.1 Detailed Description

Definition at line 69 of file data_structures.hxx.

The documentation for this struct was generated from the following file:

- /home/abhijee/Documents/GitHub/Digit_RBKD/include/digit_rbkd/rbda/data_structures.hxx

4.7 Joint Struct Reference

Public Attributes

- std::string **name**
- myint **parent_body**
- JointType **joint_type**
- Eigen::Matrix< myfloat, 3, 1 > **joint_axis**
- myfloat **damping** = 0.0
- myfloat **frictionloss** = 0.0
- myfloat **stiffness** = 0.0
- Eigen::Matrix< myfloat, -1, 2 > **limits**
- myfloat **gear_ratio** = 0.0
- myfloat **armature** = 0.0
- Eigen::Matrix< myfloat, 1, 2 > **motor_limits** = Eigen::Matrix<myfloat, 1, 2>::Zero()
- myfloat **u** = 0.0
- Eigen::Matrix< myfloat, -1, 1 > **q**
- Eigen::Matrix< myfloat, -1, 1 > **v**

Friends

- std::ostream & **operator**<< (std::ostream &os, const Joint &joint)

4.7.1 Detailed Description

Definition at line 45 of file rigid_body.hxx.

The documentation for this struct was generated from the following file:

- /home/abhijee/Documents/GitHub/Digit_RBKD/include/digit_rbkd/rigid_body.hxx

4.8 rbda::mcl_struct Struct Reference

Struct to hold mass (1x1), center of mass location (3x1), and inertia matrix (3x3).

`#include <spatial_vector_arithmetic.hxx>`

Public Attributes

- myfloat **m**
- Eigen::Matrix< myfloat, 3, 1 > **c**
- Eigen::Matrix< myfloat, 3, 3 > **I**

4.8.1 Detailed Description

Struct to hold mass (1x1), center of mass location (3x1), and inertia matrix (3x3).

Definition at line 98 of file spatial_vector_arithmetic.hxx.

The documentation for this struct was generated from the following file:

- /home/abhijee/Documents/GitHub/Digit_RBKD/include/digit_rbkd/rbda/spatial_vector_arithmetic.hxx

4.9 rbda::MotionSubspace Struct Reference

Public Member Functions

- **MotionSubspace** (JointType joint_type, Eigen::Matrix< myfloat, 3, 1 > axis)
- Eigen::Matrix< myfloat, 6, 1 > **operator*** (const Eigen::Matrix< myfloat,-1, 1 > &q)

Public Attributes

- JointType **joint_type**
- Eigen::Matrix< myfloat, 3, 1 > **axis**

4.9.1 Detailed Description

Definition at line 449 of file data_structures.hxx.

The documentation for this struct was generated from the following file:

- /home/abhijee/Documents/GitHub/Digit_RBKD/include/digit_rbkd/rbda/data_structures.hxx

4.10 rbda::MotionVector Struct Reference

Public Member Functions

- **MotionVector** (Eigen::Matrix< myfloat, 3, 1 > omega, Eigen::Matrix< myfloat, 3, 1 > v)
- **MotionVector** (Eigen::Matrix< myfloat, 6, 1 > mv)
- **MotionVector** (Eigen::Matrix< myfloat, 6,-1 > mv)
- Eigen::Matrix< myfloat, 6, 1 > **toVector** () const
- MotionVector & **operator=** (const MotionVector &rhs)
- MotionVector **cross** (const MotionVector &rhs) const
- ForceVector **cross** (const ForceVector &rhs) const

Public Attributes

- Eigen::Matrix< myfloat, 3, 1 > **omega**
- Eigen::Matrix< myfloat, 3, 1 > **v**

Friends

- std::ostream & **operator<<** (std::ostream &os, const MotionVector &mv)
- MotionVector **operator+** (const MotionVector &lhs, const MotionVector &rhs)
- MotionVector **operator-** (const MotionVector &lhs, const MotionVector &rhs)
- MotionVector **operator*** (const myfloat &lhs, const MotionVector &rhs)
- MotionVector **operator*** (const MotionVector &lhs, const myfloat &rhs)

4.10.1 Detailed Description

Definition at line 140 of file data_structures.hxx.

The documentation for this struct was generated from the following file:

- /home/abhijee/Documents/GitHub/Digit_RBKD/include/digit_rbkd/rbda/data_structures.hxx

4.11 Eigen::NumTraits< casadi::SX > Struct Reference

Public Types

- using **Real** = casadi::SX
- using **NonInteger** = casadi::SX
- using **Literal** = casadi::SX
- using **Nested** = casadi::SX

Static Public Member Functions

- static double **epsilon** ()
- static double **dummy_precision** ()
- static double **hightest** ()
- static double **lowest** ()
- static int **digits10** ()

Static Public Attributes

- static constexpr bool **IsComplex** = false
- static constexpr bool **IsInteger** = false
- static constexpr int **ReadCost** = 1
- static constexpr int **AddCost** = 1
- static constexpr int **MulCost** = 1
- static constexpr bool **IsSigned** = true
- static constexpr bool **RequireInitialization** = true

4.11.1 Detailed Description

Definition at line 73 of file common_defines.hxx.

The documentation for this struct was generated from the following file:

- /home/abhijee/Documents/GitHub/Digit_RBKD/include/digit_rbkd/common_defines.hxx

4.12 rbda::PluckerTransform Struct Reference

Public Member Functions

- **PluckerTransform** (Eigen::Matrix< myfloat, 6, 6 > X)
- **PluckerTransform** (Eigen::Matrix< myfloat, 3, 1 > euler_deg, Eigen::Matrix< myfloat, 3, 1 > p)
- **PluckerTransform** (Pose pose)
- **PluckerTransform** (Eigen::Matrix< myfloat, 3, 3 > E, Eigen::Matrix< myfloat, 3, 1 > r)
- PluckerTransform **inverse** () const
- Eigen::Matrix< myfloat, 6, 6 > **toMatrix** () const
- Pose **toPose** () const
- void **setIdentity** ()
- PluckerTransform & **operator=** (const PluckerTransform &rhs)
- ForceVector **invApply** (const ForceVector &rhs) const
- void **invApply** (Eigen::Matrix< myfloat, 6,-1 > &rhs)

Public Attributes

- Eigen::Matrix< myfloat, 3, 3 > **E**
- Eigen::Matrix< myfloat, 3, 1 > **r**

Friends

- PluckerTransform **operator*** (const PluckerTransform &lhs, const PluckerTransform &rhs)
- std::ostream & **operator<<** (std::ostream &os, const PluckerTransform &plucker_transform)
- MotionVector **operator*** (const PluckerTransform &lhs, const MotionVector &rhs)

4.12.1 Detailed Description

Definition at line 221 of file data_structures.hxx.

The documentation for this struct was generated from the following file:

- /home/abhijee/Documents/GitHub/Digit_RBKD/include/digit_rbkd/rbda/data_structures.hxx

4.13 rbda::Pose Struct Reference

Public Member Functions

- **Pose** (Eigen::Matrix< myfloat, 3, 3 > R, Eigen::Matrix< myfloat, 3, 1 > p)
- **Pose** (Eigen::Matrix< myfloat, 3, 1 > euler_deg, Eigen::Matrix< myfloat, 3, 1 > p)
- void **setIdentity** ()

Public Attributes

- Eigen::Matrix< myfloat, 3, 3 > **R**
- Eigen::Matrix< myfloat, 3, 1 > **p**

Friends

- std::ostream & **operator**<< (std::ostream &os, const Pose &pose)

4.13.1 Detailed Description

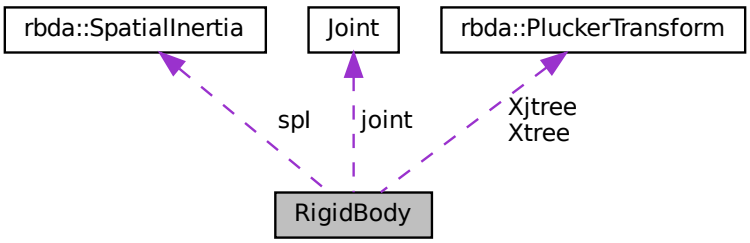
Definition at line 35 of file data_structures.hxx.

The documentation for this struct was generated from the following file:

- /home/abhijee/Documents/GitHub/Digit_RBKD/include/digit_rbkd/rbda/data_structures.hxx

4.14 RigidBody Struct Reference

Collaboration diagram for RigidBody:



Public Member Functions

- void calculate_Xjtree ()

  Returns X transform matrix after joint position q is applied.
- void **set_state** (Eigen::Matrix< myfloat, -1, 1 > q, Eigen::Matrix< myfloat, -1, 1 > v)
- void **set_pos** (Eigen::Matrix< myfloat, -1, 1 > q)
- void **set_vel** (Eigen::Matrix< myfloat, -1, 1 > v)
- void **set_u** (myfloat u)

Public Attributes

- std::string **name**
- myint **parent**
- myint **id**
- Joint **joint**
- PluckerTransform **Xtree**
- PluckerTransform **Xjtree**
- SpatialInertia **spI**

Friends

- std::ostream & **operator**<< (std::ostream &os, const RigidBody &rb)

4.14.1 Detailed Description

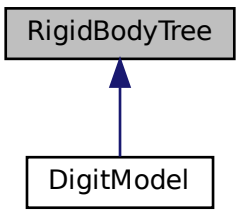
Definition at line 85 of file rigid_body.hxx.

The documentation for this struct was generated from the following file:

- /home/abhijee/Documents/GitHub/Digit_RBKD/include/digit_rbkd/rigid_body.hxx

4.15 RigidBodyTree Class Reference

Inheritance diagram for RigidBodyTree:



Public Member Functions

- void **initialize** ()
- void print_state (int precision=2, int width=25) const

  Pretty print the state of the model.
- std::vector< myint > kappa (myint body_id, myint parent_body_id=-1) const

  Get vector of ids of all bodies between body_id and parent_body_id.
- std::vector< myint > **branch_dof** (std::vector< myint > kappa) const
- PluckerTransform **spatial_body_forward_kinematics** (myint body_id, myint parent_body_id=-1) const
- MotionVector **spatial_body_velocity** (myint body_id, myint parent_body_id=-1) const
- Eigen::Matrix< myfloat, 6, -1 > **spatial_body_jacobian** (myint body_id, myint parent_body_id=-1) const
- Eigen::Matrix< myfloat, 6, -1 > **jacobian** (Pose frame, myint body_id, myint parrent_body_id=-1) const
- Eigen::Matrix< myfloat, 6, -1 > **jacobian** (Site site, myint parent_body_id=-1) const
- Eigen::Matrix< myfloat, 6, -1 > **jacobian** (myint site_id, myint parent_body_id=-1) const
- MotionVector **velocity** (Pose frame, myint body_id, myint parent_body_id=-1) const
- MotionVector **velocity** (Site site, myint parent_body_id=-1) const
- MotionVector **velocity** (myint site_id, myint parent_body_id=-1) const
- MotionVector **spatial_body_corriolis** (myint body_id, myint parent_body_id=-1) const
- Pose **forward_kinematics** (Pose frame, myint body_id, myint parent_body_id=-1) const
- Pose **forward_kinematics** (Site site, myint parent_body_id=-1) const
- Pose **forward_kinematics** (myint site_id, myint parent_body_id=-1) const
- Eigen::Matrix< myfloat, -1, -1 > **joint_space_inertia_matrix** () const
- Eigen::Matrix< myfloat, -1, 1 > **joint_space_nonlinear_effects** (bool include_gravity=true) const
- myfloat **total_mass** () const
- Eigen::Matrix< myfloat, 3, 1 > **com_position** () const
- Eigen::Matrix< myfloat, 6, -1 > **centroidal_momentum_matrix** (Eigen::Matrix< myfloat, -1, -1 > H=Eigen::Matrix< myfloat, -1, -1 >()) const
- Eigen::Matrix< myfloat, 6, 1 > **centroidal_momentum_corriolis** (Eigen::Matrix< myfloat, -1, 1 > C_↩ terms=Eigen::Matrix< myfloat, -1, 1 >()) const
- Eigen::Matrix< myfloat, -1, 1 > **get_qvec** () const
- Eigen::Matrix< myfloat, -1, 1 > **get_vvec** () const
- void **set_qpos** (const Eigen::Matrix< myfloat, -1, 1 > &q)
- void **set_qvel** (const Eigen::Matrix< myfloat, -1, 1 > &v)

Public Attributes

- std::string **name**
- myint **num_bodies** = -1
- myint **num_sites** = 0
- myint **num_q** = 0
- myint **num_v** = 0
- myint **num_u** = 0
- std::vector< RigidBody > **bodies**
- std::vector< Site > **sites**
- Eigen::Matrix< myfloat, 3, 1 > **gravity** = Eigen::Matrix<myfloat, 3, 1>(0.0, 0.0, -9.81)

4.15.1 Detailed Description

Definition at line 167 of file rigid_body.hxx.

4.15.2 Member Function Documentation

4.15.2.1 kappa()

```
std::vector<myint> RigidBodyTree::kappa (
            myint body_id,
            myint parent_body_id = -1 ) const  [inline]
```

Get vector of ids of all bodies between body_id and parent_body_id.

Parameters

body_id	
parent_body_id	

Returns

std::vector<myint>

Definition at line 217 of file rigid_body.hxx.

```
218     {
219         std::vector<myint> kappa;
220         kappa.push_back(body_id);
221         if (body_id == parent_body_id) return kappa;
222         // go up the tree
223         while (bodies[body_id].parent != parent_body_id)
224         {
225             body_id = bodies[body_id].parent;
226             kappa.push_back(body_id);
227             if (body_id == -1) return std::vector<myint>();
228         }
229         // reverse the vector
230         std::reverse(kappa.begin(), kappa.end());
231         // return the kappa
232         return kappa;
233     }
```

4.15.2.2 print_state()

```
void RigidBodyTree::print_state (
            int precision = 2,
            int width = 25 ) const  [inline]
```

Pretty print the state of the model.

Parameters

precision	
width	

Definition at line 198 of file rigid_body.hxx.

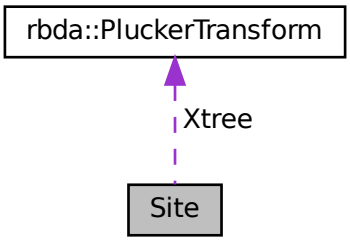
```
199     {
200         // pretty print all bodies with q and v fixed width
201         // body_name    q   v
202         std::cout « std::setw(width) « "body_name" « std::setw(width) « "q" « std::setw(width) « "v" « 
    std::endl;
203         for (int i = 0; i < bodies.size(); i++)
204         {
205             std::cout « std::setw(width) « bodies[i].name « std::setw(width) « ev2str(bodies[i].joint.q, 
    precision) « std::setw(width) « ev2str(bodies[i].joint.v, precision) « std::endl;
206         }
207 
208     }
```

The documentation for this class was generated from the following file:

- /home/abhijee/Documents/GitHub/Digit_RBKD/include/digit_rbkd/rigid_body.hxx

4.16 Site Struct Reference

Collaboration diagram for Site:



Public Attributes

- std::string **name**
- myint **parent_body**
- PluckerTransform **Xtree**

4.16.1 Detailed Description

Definition at line 37 of file rigid_body.hxx.

The documentation for this struct was generated from the following file:

- /home/abhijee/Documents/GitHub/Digit_RBKD/include/digit_rbkd/rigid_body.hxx

4.17 rbda::SpatialInertia Struct Reference

Public Member Functions

- **SpatialInertia** (myfloat m, Eigen::Matrix< myfloat, 3, 1 > c, Eigen::Matrix< myfloat, 3, 3 > Ic)
- **SpatialInertia** (myfloat m, Eigen::Matrix< myfloat, 3, 1 > c, Eigen::Matrix< myfloat, 6, 1 > Ic)
- **SpatialInertia** (Eigen::Matrix< myfloat, 6, 6 > mat)
- Eigen::Matrix< myfloat, 6, 6 > **toMatrix** () const
- SpatialInertia **apply** (PluckerTransform X) const
- SpatialInertia **invApply** (PluckerTransform X) const
- Eigen::Matrix< myfloat, 3, 1 > **com_pos** () const
- Eigen::Matrix< myfloat, 3, 3 > **com_inertia** () const

Public Attributes

- myfloat **m**
- Eigen::Matrix< myfloat, 3, 1 > **h**
- Eigen::Matrix< myfloat, 3, 3 > **I**

Friends

- SpatialInertia **operator+** (const SpatialInertia &lhs, const SpatialInertia &rhs)
- SpatialInertia **operator-** (const SpatialInertia &lhs, const SpatialInertia &rhs)
- ForceVector **operator*** (const SpatialInertia &lhs, const MotionVector &rhs)
- std::ostream & **operator<<** (std::ostream &os, const SpatialInertia &spI)

4.17.1 Detailed Description

Definition at line 336 of file data_structures.hxx.

The documentation for this struct was generated from the following file:

- /home/abhijee/Documents/GitHub/Digit_RBKD/include/digit_rbkd/rbda/data_structures.hxx

Index

add_casadi_function
    CasadiEigenCodeGen::DynamicSizeCG, 9
    CasadiEigenCodeGen::FixedSizeCG, 12

CasadiEigenCodeGen::DynamicSizeCG, 8
    add_casadi_function, 9
CasadiEigenCodeGen::FixedSizeCG, 11
    add_casadi_function, 12
    generate_eigen_function_signature, 13
    generate_eigen_function_source, 14
    get_casadi_function_io_dims, 14
    get_casadi_function_io_names, 15

DigitModel, 7

Eigen::internal::cast_impl< casadi::SX, Scalar >, 7
Eigen::NumTraits< casadi::SX >, 19

generate_eigen_function_signature
    CasadiEigenCodeGen::FixedSizeCG, 13
generate_eigen_function_source
    CasadiEigenCodeGen::FixedSizeCG, 14
get_casadi_function_io_dims
    CasadiEigenCodeGen::FixedSizeCG, 14
get_casadi_function_io_names
    CasadiEigenCodeGen::FixedSizeCG, 15

Joint, 17

kappa
    RigidBodyTree, 24

print_state
    RigidBodyTree, 24

rbda::Er_struct, 10
rbda::ForceVector, 16
rbda::mcI_struct, 17
rbda::MotionSubspace, 18
rbda::MotionVector, 18
rbda::PluckerTransform, 20
rbda::Pose, 21
rbda::SpatialInertia, 26
RigidBody, 21
RigidBodyTree, 22
    kappa, 24
    print_state, 24

Site, 25